Sun-Ting Tsai

Postdoctoral Researcher
University of Michigan, Ann Arbor
Department of Chemical Engineering
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Research Interests

I develop computational frameworks that integrate statistical mechanics, enhanced sampling, and physics-informed machine learning to predict rare-event kinetics and enable the rational design of molecular and soft-matter systems with targeted structures, dynamics, and functional behaviors.

Keywords: statistical mechanics; molecular simulations; rare-event sampling; machine learning for molecular dynamics; soft condensed matter; self-assembly; inverse design.

Selected Research Contributions

- Developed kinetic distance and reaction coordinates for rare events in molecular systems by combining enhanced sampling, spectral gap optimization, diffusion map-inspired representations, and machine learning.
- Initiated a research direction applying language-model and recurrent-neural-network ideas to molecular dynamics, leading to first-author publications on molecular trajectory learning and path sampling of neural network dynamics.
- Introduced metadynamics-based free energy exploration into hard particle Monte Carlo and digital-alchemy inverse-design simulations, extending enhanced sampling concepts to entropy-driven self-assembly.
- Built interdisciplinary collaborations across statistical physics, molecular simulation, machine learning, and quantum/computational science.

Education

Ph.D. in Physics, University of Maryland, College Park May 2022
Department of Physics and Institute for Physical Science and Technology
Advisor: Prof. Pratyush Tiwary
Co-advisor: Prof. Christopher Jarzynski
Dissertation: *Building Kinetic Models for Complex Systems with Arbitrary Memories*

M.S. in Physics, National Tsing Hua University, Hsinchu, Taiwan July 2013
Advisor: Prof. Shu-Jung Tang
Thesis: *Study of spin, electronic, and lattice structures of a quasi-topological-insulator alloy thin film*

B.S. in Physics, National Tsing Hua University, Hsinchu, Taiwan June 2011

Publications

Peer-Reviewed Journal Articles

*Author order follows the published articles. * denotes equal contribution.*

1. Charlotte Shiqi Zhao*, **Sun-Ting Tsai***, and Sharon C. Glotzer, "Exploring entropy landscapes using hard particle Monte Carlo metadynamics", *Proc. Natl. Acad. Sci.* **123**, e2537764123 (2026)
   * *Equal contribution; initiated the project, proposed the central idea, and developed the theoretical framework.*

2. Charlotte Shiqi Zhao, **Sun-Ting Tsai**, and Sharon C. Glotzer, "Hybrid Monte Carlo metadynamics (hybridMC-MetaD)", *J. Chem. Phys.* **164**, 024107 (2026)
3. Shih-Kuang Alex Lee, **Sun-Ting Tsai**, and Sharon C. Glotzer, "Classification of complex local environments in systems of particle shapes through shape symmetry- encoded data augmentation", *J. Chem. Phys.* **160**, 154106 (2024)
4. Ziyue Zou, Eric R. Beyerle, **Sun-Ting Tsai**, and Pratyush Tiwary, "Driving and characterizing nucleation of urea and glycine polymorphs in water", *Proc. Natl. Acad. Sci.* **120**, e2216099120 (2023)
5. **Sun-Ting Tsai**, Eric Fields, Yijia Xu, En-Jui Kuo, and Pratyush Tiwary, "Path sampling of recurrent neural networks by incorporating known physics", *Nature Communications* **13**, 7231 (2022)
6. Ziyue Zou, **Sun-Ting Tsai**, and Pratyush Tiwary, "Toward Automated Sampling of Polymorph Nucleation and Free Energies with the SGOOP and Metadynamics", *J. Phys. Chem. B* **125**, 13049–13056 (2021)
7. **Sun-Ting Tsai**, Zachary Smith, and Pratyush Tiwary, "SGOOP-d: Estimating Kinetic Distances and Reaction Coordinate Dimensionality for Rare Event Systems from Biased/Unbiased Simulations", *J. Chem. Theory Comput.* **17**, 6757–6765 (2021)
8. **Sun-Ting Tsai**, En-Jui Kuo, and Pratyush Tiwary, "Learning Molecular Dynamics with Simple Language Model Built upon Long Short-Term Memory Neural Network", *Nature Communications* **11**, 5115 (2020)
9. **Sun-Ting Tsai** and Pratyush Tiwary, "On the Distance between A and B in Molecular Configuration Space", *Molecular Simulation*, 1–8 (2020)
10. Li-Min Wang, **Sun-Ting Tsai**, Chih-Yu Lee, Pai-Yi Hsiao, Jia-Wei Deng, Hung-Chieh Fan Chiang, Yicheng Fei, and Tzay-Ming Hong, "Crumple-Origami Transition for Twisting Cylindrical Shells", *Phys. Rev. E* **101**, 053001 (2020)
11. **Sun-Ting Tsai**, Zachary Smith, and Pratyush Tiwary, "Reaction Coordinates and Rate Constants for Liquid Droplet Nucleation: Quantifying the Interplay between Driving Force and Memory", *J. Chem. Phys.* **151**, 154106 (2019)
12. João Marcelo Ribeiro, **Sun-Ting Tsai**, Debabrata Pramanik, Yihang Wang, and Pratyush Tiwary, "Kinetics of Ligand–Protein Dissociation from All-Atom Simulations: Are We There Yet?", *Biochemistry* **58**, 156–165 (2018)
13. Zachary Smith, Debabrata Pramanik*, **Sun-Ting Tsai***, and Pratyush Tiwary, "Multi-Dimensional Spectral Gap Optimization of Order Parameters through Conditional Probability Factorization", *J. Chem. Phys.* **149**, 234105 (2018)
14. **Sun-Ting Tsai**, Panpan Huang, Li-Min Wang, Zhengning Yang, Chin-De Chang, and Tzay-Ming Hong, "Acoustic Emission from Breaking Bamboo Chopsticks", *Phys. Rev. Lett.* **116**, 035501 (2016)
15. **Sun-Ting Tsai**, Chin-De Chang, Ching-Hao Chang, Meng-Xue Tsai, Nan-Jung Hsu, and Tzay-Ming Hong, "Power-Law Ansatz in Complex Systems: Excessive Loss of Information", *Phys. Rev. E* **92**, 062925 (2015)
16. Wei-Chuan Chen, Tay-Rong Chang, **Sun-Ting Tsai**, S. Yamamoto, Je-Ming Kuo, Cheng-Maw Cheng, Ku-Ding Tsuei, Koichiro Yaji, Hsin Lin, H.-T. Jeng, Chung-Yu Mou, Iwao Matsuda, and S.-J. Tang, "Significantly Enhanced Giant Rashba Splitting in a Thin Film of Binary Alloy", *New Journal of Physics* **17**, 083015 (2015)

Preprints and Manuscripts

1. **Sun-Ting Tsai***, En-Jui Kuo*, and Sharon C. Glotzer, "Harmonic Analysis of Point Group Symmetries in Particle Systems", Manuscript available upon request.
2. **Sun-Ting Tsai***, Haoxuan Zeng*, and Sharon C. Glotzer, "Data-driven inverse design for colloidal crystals via digital alchemy simulations", Manuscript in preparation.
3. Shih-Kuang Alex Lee, **Sun-Ting Tsai**, and Sharon C. Glotzer, "Exploratory digital alchemy for colloidal crystal discovery", arXiv:2606.13586 (2026), submitted to *npj Soft Matter*.

4. Tobias Dwyer, Hanyi Duan, Charlotte Zhao, Sumitava Kundu, Timothy C. Moore, **Sun-Ting Tsai**, Xingchen Ye, and Sharon C. Glotzer, Manuscript in preparation.

Selected Presentations

Invited Seminars

- "Evaluating local point group symmetry via Rayleigh quotients," Center for Simulation and Data Intensive Research (CSiDIR) Seminar, University of Michigan, Ann Arbor, MI, 2025.

Contributed Oral Presentations

- "Metadynamics with Non-Differentiable Collective Variables," AIChE Annual Meeting, Boston, MA, 2025.
- "Classification of Complex Local Environments in Systems of Particle Shapes through Shape-Symmetry-Encoded Data Augmentation," MRS Meeting, Boston, MA, 2024.
- "Learning Molecular Dynamics with Simple Language Model Built upon Long Short-Term Memory Neural Network," APS March Meeting, 2021.

Poster Presentations

- "Dynamic Disorder and Rare Event Kinetics," Gordon Research Conference, Holderness, New Hampshire, 2018.

Grant-Writing and Funded Research Impact

- Contributed data, figures, and methodological text to NSF-funded grant proposal on self-assembly pathway study; proposal funded, PI: Prof. Sharon C. Glotzer.
- Results from my collaborative research leading to two *Nature Communications* publications subsequently served as preliminary data supporting a successful research grant awarded by Taiwan's National Science and Technology Council (NSTC); PI: Prof. En-Jui Kuo.
- The research work leading to the *Phys. Rev. Lett.* publication while I was a research assistant at NTHU contributed preliminary results and figures supporting a successful grant awarded by Taiwan's National Science and Technology Council (NSTC), PI: Prof. Tzay-Ming Hong.

Professional Service

Ad Hoc Manuscript Reviewer

- *Nature Machine Intelligence*, 2025.
- *Journal of Chemical Theory and Computation*, 2 manuscripts, 2023, 2025.
- *Journal of Chemical Physics*, 2023.

Research Experience

Postdoctoral Researcher, University of Michigan, Ann Arbor 2022–Present
Advisor: Prof. Sharon C. Glotzer

- Introduced metadynamics-based enhanced sampling to hard particle Monte Carlo simulations, enabling the exploration of entropy-driven self-assembly that were difficult to access with conventional sampling.

- Integrated metadynamics and machine learning with inverse-design simulations to improve characterization and optimization of self-assembling particle systems.
- Developed a representation-theoretic measure for continuous quantification of point group symmetry in particle systems across structural scales.
- Developed machine learning approaches for classifying complex local particle environments using symmetry-informed data augmentation.
- Outcomes: publications in *PNAS*, *JCP*, and related journals.

Graduate Research Assistant, University of Maryland, College Park 2017–2022
Advisor: Prof. Pratyush Tiwary

- Developed computational methods for rare event dynamics, reaction coordinate discovery, and kinetic distance estimation.
- Initiated and developed a research direction applying recurrent neural networks and language-model concepts to molecular dynamics, overcoming challenges in learning long molecular trajectories.
- Established collaborations with researchers in quantum and computational science to develop physics-informed path-sampling approach for recurrent neural networks.
- Integrated enhanced sampling, spectral gap optimization, and machine learning to study nucleation, memory effects, and molecular kinetics.
- This work resulted in first-author publications in *Nature Communications*, *JCTC*, *JCP*, and related journals.

Research Assistant, Soft Matter Physics Laboratory at National Tsing Hua University 2014–2017
Advisor: Prof. Tzay-Ming Hong

- Investigated fracture, crumpling, and mechanical instabilities in soft-matter systems using simulations, experiments, and statistical modeling.
- Developed molecular simulation and data-analysis approaches for modeling plastic deformation and acoustic-emission statistics.
- Helped lead group research discussions and collaborative data-analysis efforts on fracture, crumpling, and complex-systems projects.
- Resulted in first-author publications in *Physical Review Letters* and *Physical Review E*.

Graduate Researcher, Surface Science Lab and NSRRC, National Tsing Hua University 2011–2013
Advisor: Prof. Shu-Jung Tang

- Investigated spin, electronic, and lattice structures of quasi-topological-insulator alloy thin films using ARPES, spin-resolved ARPES, and LEED.
- Conducted synchrotron-based measurements at NSRRC, Taiwan, and Photon Factory, KEK, Japan.
- Contributed to a publication in *New Journal of Physics*.

Mentoring and Research Supervision

University of Michigan, Ann Arbor 2022–Present

- As a postdoc in the Glotzer Group, I have served as day-to-day research mentor for four Ph.D. students and one undergraduate student on projects involving enhanced sampling, inverse design, self-assembly, and machine-learning-based analysis of particle systems.
- **Charlotte Shiqi Zhao**, Ph.D., University of Michigan, 2025. Co-mentored research on metadynamics-based enhanced sampling for entropy-driven self-assemblies from colloidal particle shapes; outcomes include two peer-reviewed publications in *J. Chem. Phys.* and *PNAS*.

- **Alex Shih-Kuang Lee**, Ph.D., University of Michigan, 2025. Co-mentored research on symmetry-informed machine-learning methods for classifying complex local environments in particle systems; outcomes include one peer-reviewed publication in *J. Chem. Phys.*. And one submitted preprint on exploratory digital alchemy for colloidal crystal discovery.
- I currently co-mentor two Ph.D. students and one undergraduate student on computational inverse design methods; one manuscript in preparation.
- I led a methods-development subgroup focused on metadynamics, collective-variable design, and enhanced sampling, coordinating research discussions and technical development with graduate-student collaborators.

Teaching Experience

Guest Lecturer for a Graduate-Level Chemical Engineering Course taught by Assistant Professor Rebecca Lindsey 2024
University of Michigan, Ann Arbor

- Delivered a lecture on enhanced sampling methods, with emphasis on metadynamics and applications to molecular simulation.
- Independently designed and led a hands-on computational module demonstrating metadynamics through example code and simulation analysis.

Graduate Teaching Assistant, Undergraduate General Physics Laboratory 2017–2018
University of Maryland, College Park

- Led weekly laboratory instruction and discussion for undergraduate students in introductory physics.
- Guided students in experimental procedures, data analysis, uncertainty estimation, and scientific reasoning.
- Evaluated lab reports and provided feedback on experimental reasoning and quantitative analysis.

Teaching Assistant, Undergraduate Quantum Physics 2012
National Tsing Hua University, Hsinchu, Taiwan

- Led recitation and problem-solving sessions for undergraduate quantum physics.
- Assisted students with conceptual understanding, exam preparation, and exam review during office hours.

Technical Skills

Simulation and modeling: Molecular dynamics, rare-event sampling, metadynamics, SGOOP, reaction-coordinate discovery, stochastic dynamics, statistical model selection.

Programming and software: Python, C/C++, MATLAB, Mathematica, R, LAMMPS, HOOMD-blue, Gromacs, PLUMED, PyTorch,TensorFlow, Igor Pro.

Experimental methods: ARPES, spin-resolved ARPES, LEED, synchrotron-based measurements, acoustic emission analysis.

Additional Experience

Military Service 2013–2014